MDAS Meter-Modem Automation: User Guide

Two scripts fill the MDAS Meter-Modem form from an Excel sheet: `mdas_auto.py` (Any-SIM) and `mdas_auto_own_sim.py` (Own-SIM).

What they do

For each meter the script enters the meter and modem, finds a SIM that works on the site, fills date, project, town and transformer, clicks Save, and writes the result to Excel. You only log in at the start.

Which one to use

	Any-SIM	Own-SIM
File	`mdas_auto.py`	`mdas_auto_own_sim.py`
SIMs tried	The meter's own two, then SIMs from other rows	The meter's own two only
No SIM works	SIM NOT AVAILABLE / ALREADY USED	NO OWN SIM MATCHED
Result file	`MDAS_result.xlsx`	`MDAS_result_own_sim.xlsx`

Use the current `mdas_auto.py`. For own SIMs only, set `USE_SIM_POOL = False` in it.

Setup (once)

- Install Python 3, then run `pip install playwright pandas openpyxl` and `python -m playwright install chromium`.
- Put the script and `MDAS_pendding.xlsx` in one folder.
- Close the result Excel before running.

Input Excel columns

Column	Required	Notes
Meter Sr. No.	Yes	For example X2073610
Circle Name	Yes	Used as the Town
New Meter Installation Date	Yes	If blank, the meter is skipped
Sim No	No	First SIM to try
Optional Sim No	No	Second SIM. The column may be missing

Letters in a SIM (such as a trailing U) are removed. The modem number is built as M + meter number, so a Modem IMEI column is ignored.

How to run

1. At the top of the script set `DRY_RUN = True` and `LIMIT = 3` (test mode, nothing is saved).
2. In the folder, run `python mdas_auto.py` (or `python mdas_auto_own_sim.py`).
3. Log in in the browser, stay on the home page, press ENTER in the terminal.
4. Check the result Excel. If it is right, set `DRY_RUN = False`, run 10 meters (`LIMIT = 10`), then `LIMIT = None` for all.
5. If the site logs you out, log in again and press ENTER. Stop with Ctrl+C. Re-running skips meters already SAVED or DONE.

Result columns and status

Column	Meaning
Sim Used	The SIM that worked
IP / Telephone	Returned for that SIM
Assigned To	Transformer chosen
Status	See below
Message	SIMs that failed, and why

Status	What to do
SAVED	Nothing. Saved on the site.
DONE (already entered)	Nothing. It was already on the site.
SIM NOT AVAILABLE / ALREADY USED, NO OWN SIM MATCHED	No usable SIM. The meter needs a new SIM.
SUBDIVISION NOT FILLED, SAVE ERROR, CHECK	Read Message and check the meter on the site.
MISSING INSTALL DATE, TOWN NOT FOUND	Fix the Excel sheet.
SCRIPT ERROR	See Message and `error_<meter>.png`.

Cautions

- Save writes real data to MDAS. Test first.
- Any-SIM can give a meter a SIM from another row. Review rows where Sim Used shows (POOL.
- Never share your MDAS password.

Quick fixes

Problem	Fix
`playwright` is not recognized	Run `python -m playwright install chromium`
PermissionError on the result file	Close it in Excel. The script writes `MDAS_result_live.xlsx` meanwhile
A SIM keeps being skipped	Delete `linked_sims.json` (Any-SIM) or `bad_sims_own_sim.json` (Own-SIM)
Saved SIM for an old run	Run `python find_saved_sim.py`